

STRATEGIC ASSESSMENT:-

Strategic assessment:

in ~~the~~ project evaluation is the process of evaluating a project's alignment with the goals and objectives of an organization.

Key Elements of Strategic assessment

- i) Alignment with organizational goals.
- ii) Contribution in Competitive Advantage.
- iii) fulfills stakeholders needs.
- iv) Portfolio fit. → should fit in current eco-system.
- v) Long-Term Impact.

Typical issues and questions to be considered during strategic assessment

IS → Information system.

Objective: How will the proposed system contribute to the organization's objectives.
→ hardware, software etc (Basically a system)

IS plan: How does the proposed system fit into IS plan? How will it interact with systems proposed for later development.

Organization structure: what effect will the new system have on the existing organization structure.

→ summarized/detailed/real time daily/weekly/monthly

MIS: what information will the system provide and at what level in the organization? In what way will it enhance existing management IS.

Personnel: In what way will the proposed system affect ~~the~~ manning levels and the existing employee skill base.

no. of emp required.

Image: what, ~~if any~~, will be the effect on customers attitudes towards the organization.

IS → Information system

MIS → Management information system.

Cost-benefit analysis ↓

- Identifying and estimating all of the costs and benefits of carrying out the project.
- This include development costs of the system, the operating costs and the benefits from the system.
- Expressing these costs and benefits in common units.

* Categorizing cost according to where they originate in project.

i. Development costs - include the salaries and other employment costs of the staff. of the employees

ii. Setup costs - include costs of putting system into place.

iii. Operational costs - consist the costs of operating the system once it has been installed.

* Benefits can be categorized as:

i. Direct benefits: These accrue directly from the operation of the proposed system.

ii. Assessable indirect benefits: these are generally secondary benefits, that arise as a by-product of the project.

iii. Intangible benefits: These are long term benefits which contribute to the organization's success.

Cost-benefit evaluation techniques

i) Net profit:

The net profit of a project is the difference between the total costs and the total income over the life of the project.

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	15,000
Net Profit	50,000 50,000	100,000 100,000	50,000	75,000

ii) Payback period:

is the time taken to break even or pay back the initial investment.

$$\text{Payback period} = \text{Initial Investment} / \text{Yearly cash flow}$$

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	15,000
Payback	5 years	5 years	3.33 years	4 years

iii) Return on investment (ROI) / accounting rate of return.

Also known as accounting rate of return (ARR), provides a way of comparing the net profitability to the investment required.

$$ROI = \frac{\text{average annual Profit}}{\text{total investment}} \times 100 \quad \text{or } NP/\text{Number of years}$$

Q. Calculate ROI for each project in the table and decide which one must be chosen.

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000

i) Project I

$$ROI = \frac{\text{average annual Profit}}{\text{total investment}} \times 100 = \frac{10,000}{100,000} \times 100 = 10\%$$

ii) Project II

$$ROI = \frac{20,000}{1,000,000} \times 100 = 2\%$$

iii) Project III

$$ROI = \frac{10,000}{100,000} \times 100 = 10\%$$

iv) Project IV

$$ROI = \frac{15,000}{120,000} \times 100 = 12.5\%$$

∴ Project IV has highest return on investment so it must be chosen.

iv. Net Present value (NPV):

(5)

The calculation of net present value is a project evaluation technique that takes into account the profitability of project and the timing of cash flows that are produced.

$$\text{Present value} = \frac{\text{value in year } t}{(1+r)^t} \quad \text{r-discounted rate}$$

- Discount factor = $\frac{1}{(1+r)^t}$
- NPV = discount factor $\times$ value in year t

Q. Assuming 10% discount rate calculate the NPV of all projects.

Project I	Year	Project 1	$\frac{1}{(1+10\%)^t}$	Discounted cash flow
	0	-100,000	1.0000	-100,000
	1	10,000	0.9091	9091
	2	10,000	0.8264	8264
	3	10,000	0.7513	7513
	4	20,000	0.6830	13660
	5	100,000	0.6209	62090
	Net profit	50,000	NPV	61811

Project II

Year	Project 2	$\frac{1}{(1+10\%)^t}$	Discounted cash flow
0	-1,000,000	1.0000	-1,000,000
1	200,000	0.9091	181820
2	200,000	0.8264	165280
3	200,000	0.7513	150260
4	200,000	0.6830	136600
5	300,000	0.6209	186270
Net profit	100,000	NPV	-179770//

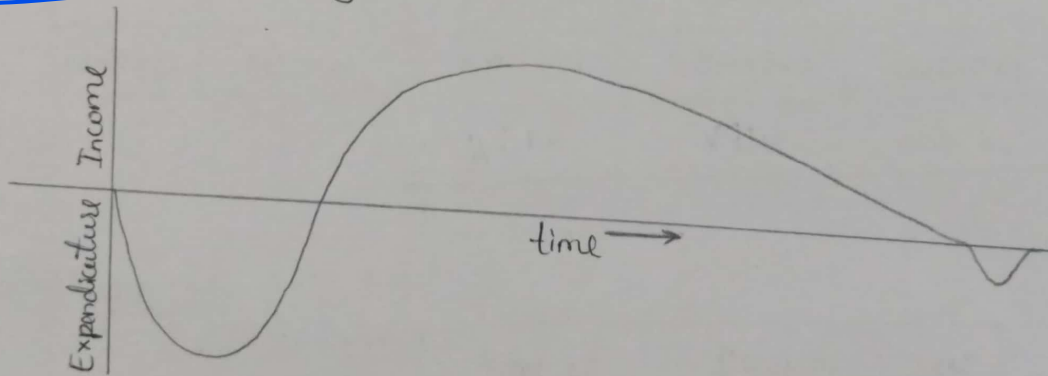
Project III & IV

Year	Project 3	Project 4	$\frac{1}{(1+0.1)^t}$	Discount rate flow Project 3	Discount rate flow Project 4
0	-100,000	-120,000	1.0000	-100,000	-120,000
1	30,000	30,000	0.9091	27273	27273
2	30,000	30,000	0.8264	24792	24792
3	30,000	30,000	0.7513	22539	22539
4	30,000	30,000	0.6830	20490	20490
5	30,000	75,000	0.6209	18627	46567.5
Net profit	50,000	75,000	NPV	13721	21661.5

① $3 > 4 //$

choosing project I will be beneficial.

* cash flow forecasting



Development stage: High cash outflows due to research, design, testing and initial production setup.

Introduction stage: Continued high cash outflows for marketing, distribution setup.

Growth stage: Cash inflows increase significantly due to growing sales and market adoption.

Maturity stage: Cash inflows peak as the market saturates and sales growth slows.

Decline stage: Cash inflow decreases as demand falls due to market saturation and competition.

Risk evaluation

→ not completed

- * Project risks: prevents the project from being completed successfully.
- * Business risks: prevents the project from being profitable.

→ not profitable.

is Risk identification and ranking

- Constructing project risk matrix utilizing a checklist of possible risks and classify according to its relative importance and likelihood. (low, high, medium).

Risk	Importance	likelihood
• Client reject proposed look and feel of site	H	—
• Competitors undercut price	H	M
• Warehouse unable to deal with increased demand	M	L
• online payment has security problems	H	M
• Maintenance costs higher than estimated	L	L
• Response time deter purchasers	M	M

e-commerce application

ii) Risk and net present value

- If project is relatively risky, then use higher discount rate to calculate NPV.
- Reasonably safe project → additional 2% risk premium.
- Fairly risky project → additional 5% risk premium.
- Risks are generally categorized using scoring method... and risk premiums can be designated for each category some means of considering risk.

iii) Cost-benefit analysis (Decision tree types).

- It is required to consider each possible outcome.
- Estimate the probability of its occurring and the corresponding value of the outcome.
→ instead of one cash flow for whole project → do cash flow for each outcome.
- Instead of single cash flow, consider set of cash flow forecasts.
- The value of the project is then obtained by summing the cost or benefit for each possible outcome.

iv) Risk profile analysis

By studying the results of a sensitivity analysis we can identify those factors that are most important to the success of the project.

v) using decision tree:

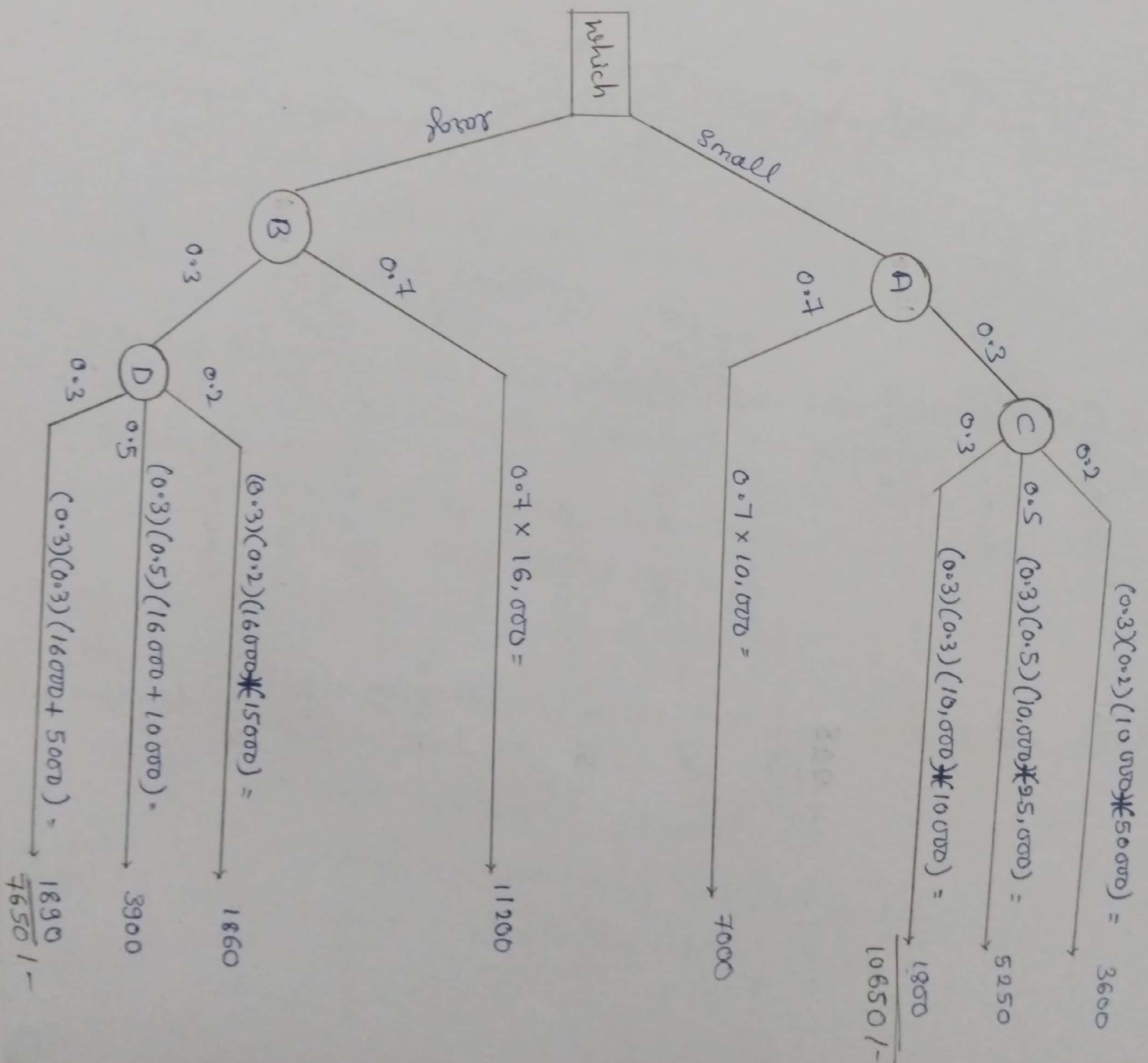
helps in risk evaluation in an effective way to systematically analyze and visualize risks and uncertainties in a project.

- Risk identification & ranking
- Risk & NPV
- cost-benefit analysis → risk profile analysis.
- using decision tree

You are planning an activity which requires the use of a crane (9)
 30% probability that the activity will become critical.

Delay	Small crane		Large crane	
	Probability	Delay in days	Probability	Delay in days
Long	0.2	10	0.2	3
medium	0.5	5	0.5	2
short	0.3	2	0.3	1

Small crane cost - 10,000 Large crane cost - 16,000
 each days delay = 5000

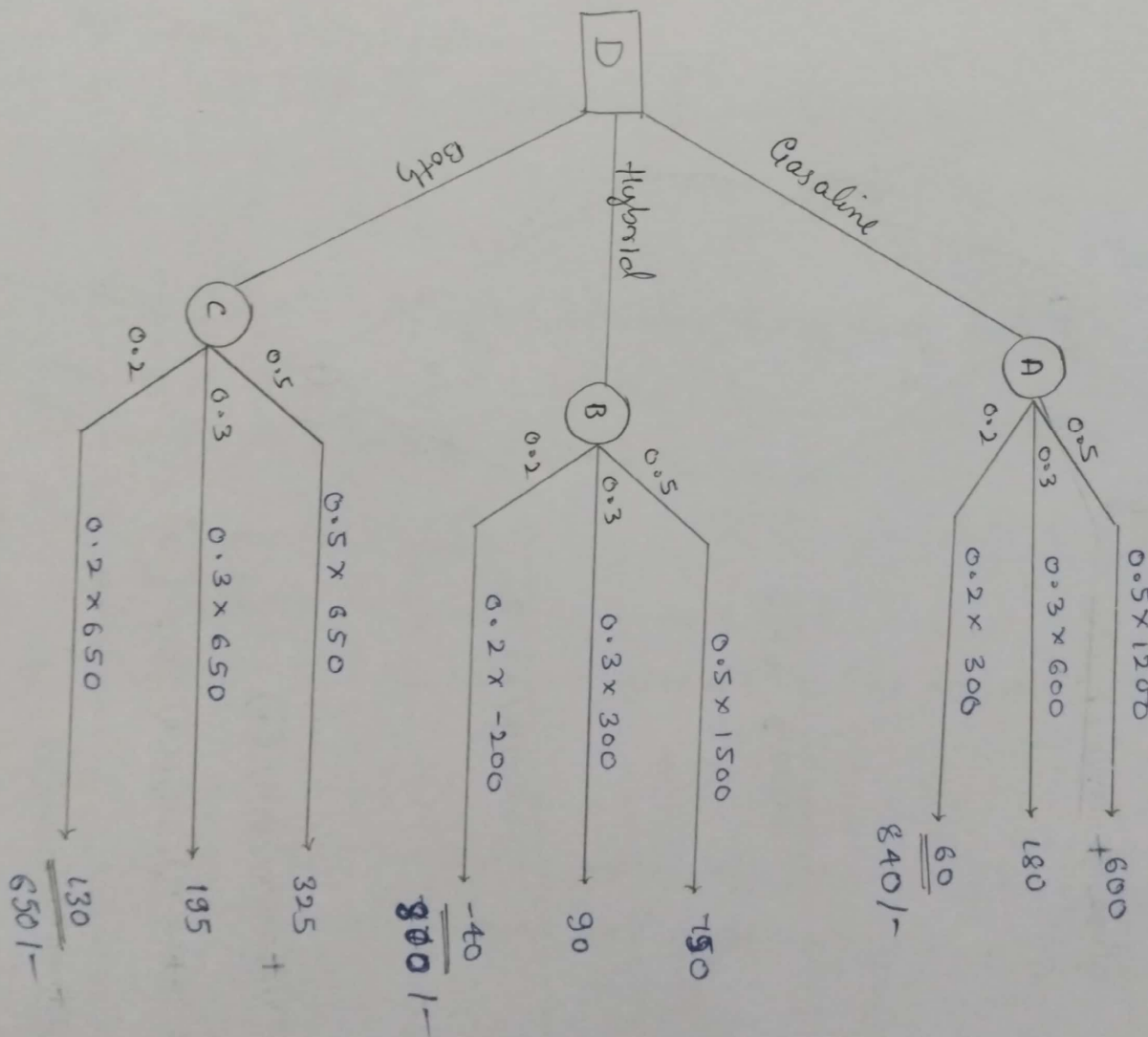


Gas price	Probability	Profit		
		Gasoline	Hybrid	Both
Increase	0.5	1200	1500	650
Decrease	0.3	600	300	650
Fluctuate	0.2	300	-200	650

$$EMV(\text{Gasoline}) = (0.5 \times 1200) + (0.3 \times 600) + (0.2 \times 300) = 840$$

$$EMV(\text{Hybrid}) = (0.5 \times 1500) + (0.3 \times 300) - (0.2 \times 200) = 800$$

$$EMV(\text{Both}) = (0.2 \times 650) + (0.3 \times 650) + (0.5 \times 650) = 650$$



Various kinds of risk in software development

i> Schedule risk: Time related risk or project delivery risks.
Some reasons are:

- Time is not estimated perfectly.
- Improper resource allocation.

ii> Budget Risk: It mainly occurs due to budget overruns.

- Improper budget estimation.
- Mismanagement in budget handling.
- Improper tracking of Budget.

iii> Operational Risks: These are the risks which happen in day-to-day operational activities during project development due to improper implementation.

- No proper planning about project.
- Less number of skilled people.
- Insufficient training.

iv> Technical Risks: Functional risk or performance risk which is mainly associated with functionality of product or performance part of the software product.

- Less use of future technologies.
- Less number of skilled employees.
- Improper integration of modules.

v> External Risks: Unavoidable in nature. These risks come from outside and it is out of control.

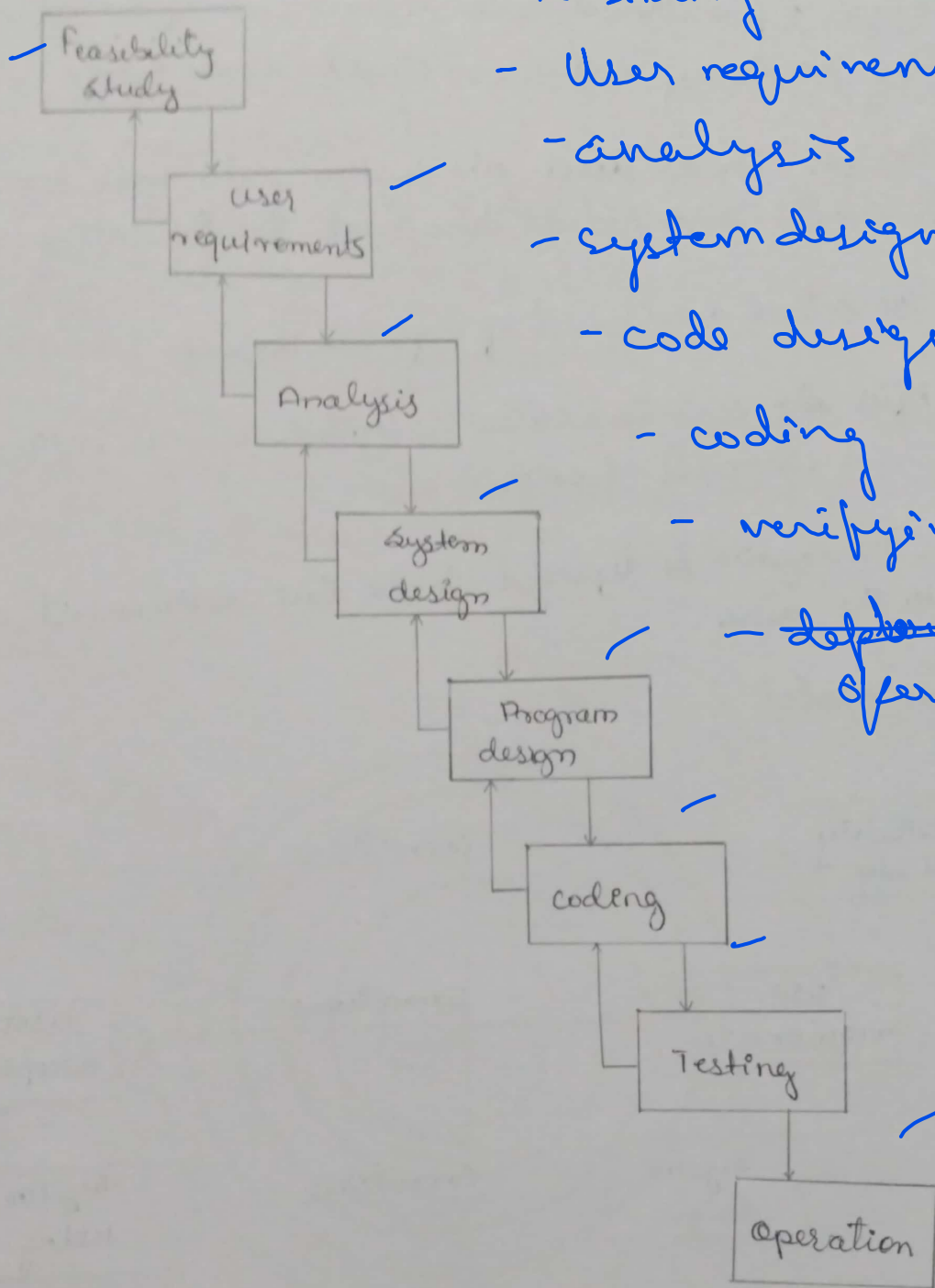
- Rapid development of market.
- Changes in Government rules/policy.
- Loss of contracts due to any reason.

~~The water fall model~~

Technical plan contents list:

1. Introduction and summary of constraints.
 - a) Character of system to be developed.
 - b) Risks and uncertainties of the project.
 - c) User requirements concerning implementation.
2. Recommended approach
 - a) Selected methodology or process model.
 - b) Development methods.
 - c) Required software tools.
 - d) Target hardware / software environment.
3. Implementation
 - a) Required development environment.
 - b) Required maintenance environment.
 - c) Required training.
4. Implications
 - a) Project products and activities - these will have an effect on the schedule duration and overall project effort.
 - b) Financial - this report will be used produce costings.

Waterfall model



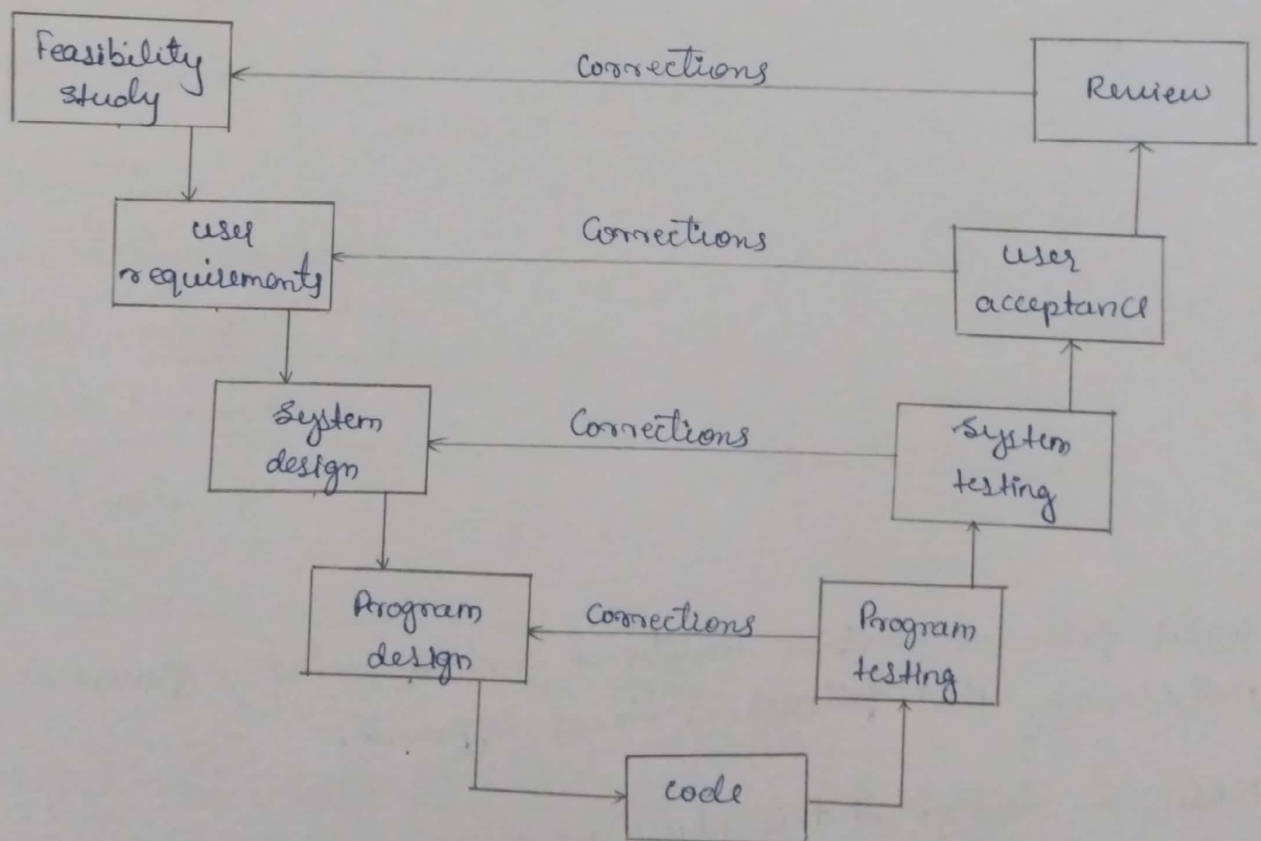
- Feasibility test
- User requirements
- analysis
- system design
- code design
- coding
- verifying & testing
- ~~deployment~~ operation.

water fall model is a traditional and sequential software development methodology that follows a linear approach.

- Feasibility Study: In this phase we determine if the project is technically, economically and operationally feasible.
- User Requirement: Focuses on gathering detailed functional and non-functional requirements from the end users.
- Analysis: Requirements collected in the previous phase are analyzed to a structure view of the system.

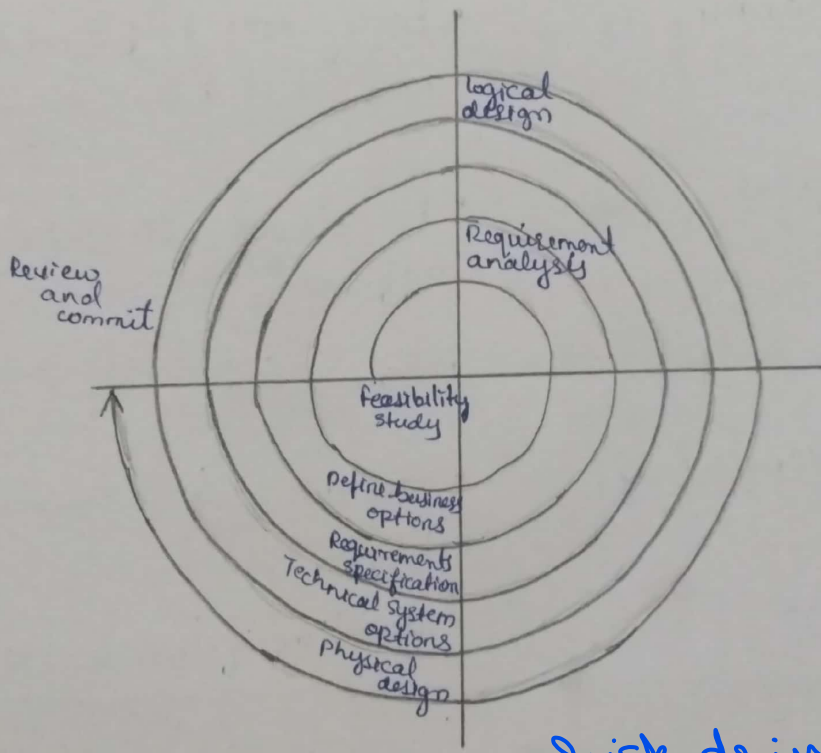
- iv) System Design: Define the overall architecture of the system, including hardware, software, network and data.
- v) Program Design: Break down the system into modules or smaller units and act as blueprint for the coding phase.
- vi) Coding: The actual development of software begins.
- vii) Testing: After the code is written, all modules are integrated and tested as a complete system.
- viii) Operation: The system is deployed in the live environment and handed over to the users.

The V-process Model



The V-process Model, also known as the verification and validation model is a software development methodology that extends the waterfall Model by emphasizing testing (validation) at every stage of the development process. It is called "V" model because of its shape, where left side represents the development phases (^{left} verification) and the right side represents the corresponding testing phases (^{right} validation). Each phase on the left-hand side has a corresponding testing phase on the right-hand side to ensure that the system meets its requirements.

Spiral model :



risk driven

The Spiral model is a risk-driven software development model that combines the iterative nature of prototyping with the systematic approach of the waterfall model. It is well-suited for large, complex, and high-risk project. The spiral model continues risk assessment, iterative refinement and incremental development across multiple phases of "spirals".

Difference between waterfall & Spiral model

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Waterfall Model	Spiral Model
• Simple and easy • Works in sequential method • Errors or risks are identified and rectified after the completion of stages. • Suitable for small projects • Early stage planning is necessary • Flexibility to change is difficult • High risk • Customer involvement is minimum • Linear framework	• Lot more complex • Works in evolutionary method • Errors or risks are identified and rectified earlier • Suitable for large projects • Early stage planning is not that necessary. • Flexibility to change is not difficult. • Low risk • Customer involvement is high • Iterative framework